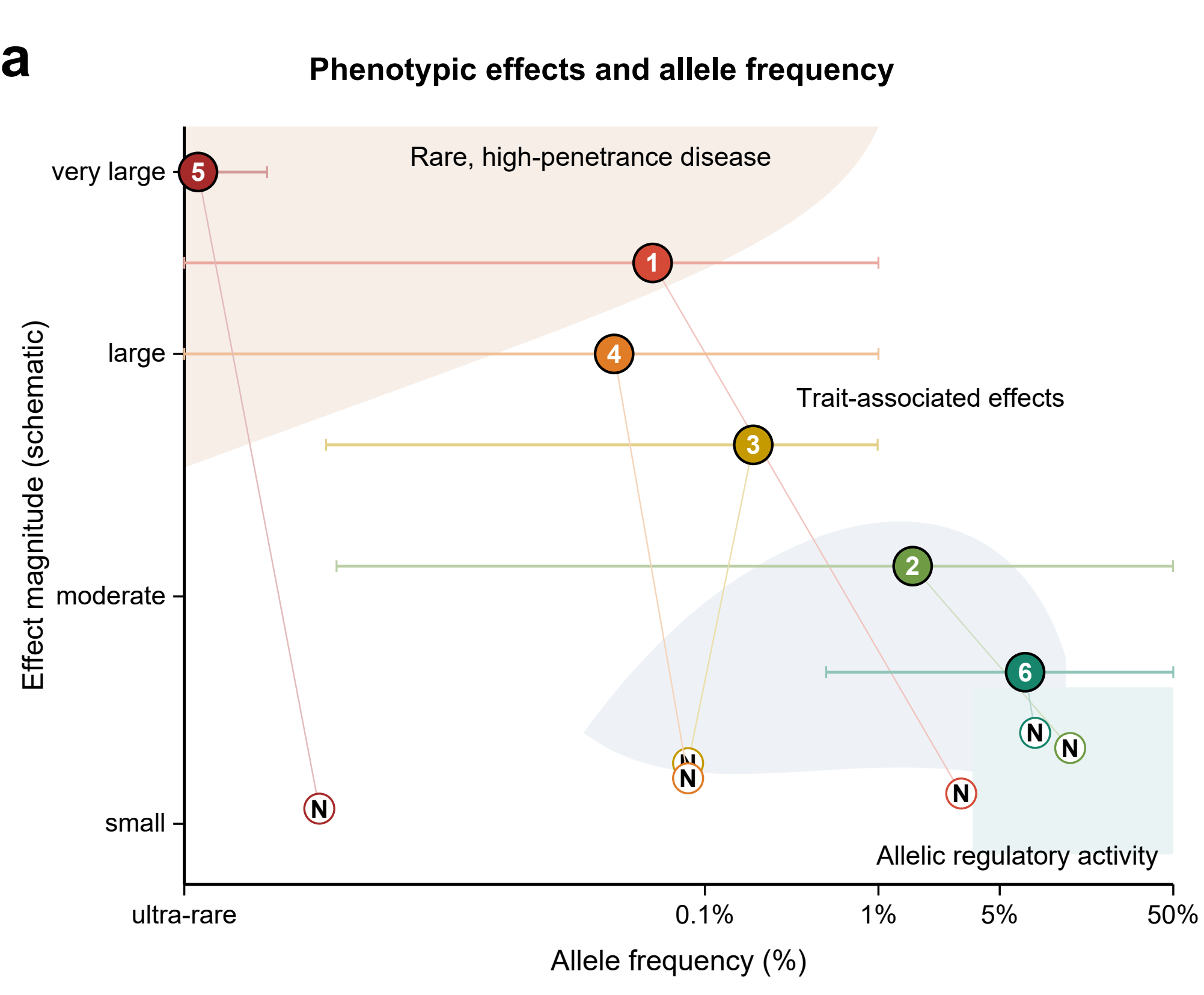

a
Phenotypic effects and allele frequency
Rare, high-penetrance disease
Trait-associated effects
Allelic regulatory activity
very large
large
moderate
small
Effect magnitude (schematic)
5
1
4
3
2
6
N
N
N
N
N
N
ultra-rare
0.1%
1%
5%
50%
Allele frequency (%)